3-Mark Questions & Answers

Q1. Explain the one-address and two-address instruction formats with examples.

1. One-Address Instruction Format:

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**Accur

2. Two-Address Instruction Format:

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Q2. Explain the fetch cycle with register transfer operations.

The Fetch Cycle reads the next instruction opcode from memory into the CPU using register transfer steps:

1. `MAR <- PC` (Copy address from Program Counter to Memory Address Register).

2. `MDF

Q3. Differentiate between hardwired and microprogrammed control unit (any 3 points).

| Feature | Hardwired Control Unit | Microprogrammed Control Unit |

| :--- | :-

Q4. Explain the microinstruction format with its three fields.

A Microinstruction Format consists of three primary fields:

```
text
```

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1. Control Signals Field: Binary bits that directly activate micro-operations (ALU lines, bus gates).

2. Conc

Q5. Explain bit-level and instruction-level parallelism with examples.

1. Bit-Level Parallelism: Increasing the word size of the processor (e.g., upgrading from 16-bit to 64-bit). Operates on 64 bits simultaneously in a single ALU cycle rather than multiple 16-bit steps.

2. Instru
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Q6. Explain SISD and SIMD architecture with examples (Flynn's classification).

1. SISD (Single Instruction, Single Data):

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Q7. Explain any two pipeline hazards with examples.

1. Data Hazard (RAW): Occurs when an instruction depends on the result of a preceding instruction that has not yet completed write-back.

* *Exa

Q8. Explain the interrupt processing cycle briefly.

When an Interrupt is acknowledged at the end of an instruction cycle:

1. Save CPU Context: Pushes current Flags, Code Segment (`CS`), and Instruction Pointer (`IP`) onto the stack.

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Q9. Explain the basic 5-stage instruction pipeline (IF, ID, EX, MEM, WB).

1. IF (Instruction Fetch): Fetch instruction from memory using PC.

2. ID (Ir

Q10. State the advantages and disadvantages of a microprogrammed control unit.

* **Advantages:**

1. High

Q11. Explain data hazards and their types (RAW, WAR, WAW).

Data hazards occur due to operand dependencies between overlapping instructions:

1. RAW (Read-After-Write / True Dependency): Inst 2 reads an operand before Inst 1 writes it.

2. WAR

Q12. Explain why parallel processing is required (any three reasons).

1. Overcoming Physical Limits of Clock Speed: Physical heat dissipation and power consumption limit raw clock rate scaling.

2. Incr
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